SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

CO1-AT2-Constraint-Based Problem Solving

Register Number	**192412678**
Name	**L Swetha**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I _____L Swetha____ (192412678) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:L Swetha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

ANSWERS

1)A) Problem statement

A hospital must assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) while satisfying all the given constraints. Backtracking Search assigns shifts one at a time and checks whether each assignment is valid. If a constraint is violated, it backtracks and tries another assignment.

Step 1: Define the variables and domains

- Variables: D1, D2, D3
- Domain: Morning (M), Afternoon (A), Night (N)

Step 2: Apply the constraints

- D1 cannot work Night.
- D2 must work before D3, meaning D2's shift must occur earlier than D3's shift.
- Only one doctor can work each shift.
- D3 cannot work Morning.
- Each doctor must receive exactly one shift.

Step 3: Backtracking Search with constraint checks

Step	Assignment	Constraint check	Result
1	D1 = Morning	D1 is not assigned Night	Valid
2	D2 = Afternoon	D2 must work before D3	Continue
3	D3 = Night	D2 (Afternoon) is before D3 (Night); D3 is not Morning	Valid
4	Check all shifts	Each shift has one doctor; all doctors are assigned	Success

Final valid schedule

Doctor	Assigned shift
D1	Morning
D2	Afternoon
D3	Night

Conclusion: The schedule satisfies all five constraints. Backtracking Search finds a valid assignment by checking each constraint before proceeding to the next variable.

2)A) Robot Navigation in a 5×5 Grid Using Manhattan Distance Heuristic

Problem Statement

A robot must move from the Start (S) to the Goal (G) in a 5×5 grid. It can move Up, Down, Left, or Right. Obstacles block certain cells, and each movement has a cost of 1.

1. Manhattan Distance Formula

$h(n)=|xn-xG|+|yn-yG|$

Where:

(xn,yn): Current cell coordinates.

(xG,yG): Goal cell coordinates.

(h(n)): Estimated distance to the goal.

2. Search Algorithm: A*

The A* algorithm uses Manhattan Distance to guide the robot toward the goal.

[$f(n)=g(n)+h(n)$]

- (g(n)): Actual cost from Start to the current cell.
- (h(n)): Estimated cost from the current cell to Goal.
- (f(n)): Estimated total path cost.

3. Steps Involved

- Start at S with an initial cost of 0.
- Calculate the Manhattan Distance from S to G.
- Explore the neighbouring cells: Up, Down, Left, and Right.
- Avoid obstacles and update the cost of valid neighbouring cells.
- Calculate ($f(n)=g(n)+h(n)$) for each candidate cell.
- Select the cell with the lowest (f(n)) and continue searching.
- Repeat until the goal G is reached.

4. Example of an Optimal Path

Since the exact obstacle positions are not provided, consider this illustrative path:

S → (2,1) → (3,1) → (3,2) → (3,3) → (3,4) → (3,5) → (4,5) → G

The path contains 8 moves.

[text{Total Cost}=8\times1=\boxed{8}]

Conclusion

A* search uses Manhattan Distance to guide the robot toward the goal while avoiding obstacles. With four-directional movement and a movement cost of 1, Manhattan Distance is an admissible heuristic, so A* can find an optimal path. The exact path and cost depend on the actual positions of the start, goal, and obstacles.

3)A) Autonomous Rescue Robot Using AI Search Strategies

a) Analysis of Constraints

- Movement: The robot can move up, down, left, or right.
- Obstacles: Blocked cells must be avoided.
- Limited sensing: The robot can observe only adjacent cells.
- Movement cost: Each normal move costs 1.
- Risky zones: Entering a risky zone costs \(1+2=3\).
- Dynamic knowledge: The robot updates its map as it discovers new information.
- Safety: It must minimize cost while avoiding dangerous paths.

b) Solution Using UCS and Online Search

An Online Search Agent combined with Uniform Cost Search (UCS) is suitable.

Steps:

- Start at S and observe nearby cells.
- Identify obstacles, safe paths, and risky zones.
- Assign cost 1 to normal moves and cost 3 to risky moves.
- Use UCS to select the lowest-cost known route.
- Move one step and observe the surroundings.
- Update the map and replan if new obstacles are discovered.
- Continue until the robot reaches G

[text{Total Cost}=\text{Normal Moves}+3(\text{Risky Moves})]

c) Application of AI Concepts

1. Intelligent Agent: The robot uses sensors to observe its surroundings and takes actions to reach the survivor.

2. Rationality: It selects the safest, lowest-cost route based on the information available and changes its plan when necessary.

3. Environment Type:

- Partially observable: Only adjacent cells are visible.
- Sequential: Each movement affects future decisions.
- Dynamic: The robot updates its knowledge during exploration.
- Discrete: Movement occurs between grid cells.
- Uncertain: Hidden obstacles and changing conditions may affect navigation.

d) Why Is This Approach Suitable?

UCS finds the least-cost route by considering normal movement costs and additional risk costs. Online search allows the robot to explore an unknown environment, update its map, avoid obstacles, and replan when new information becomes available.

Conclusion

An online search agent combined with UCS helps the rescue robot reach the survivor with minimum cumulative cost while adapting to unknown obstacles and prioritizing safe navigation.